

Supplementary Information: Data Quality Assurance Analysis Keheala Study 2

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1 Patient Characteristics

Table 1 compares patient characteristics across three cohorts: all patients registered in the TIBU electronic system during the study period, patients at study clinics recorded in TIBU (MITT sample), and patients at study clinics whose records were audited in the paper registry DQA.

Table 1: Patient characteristics across TIBU and paper registry cohorts.

Characteristic	All of TIBU (N=119,653)	Study clinics, TIBU (N=15,487)	Study clinics, paper (N=3,370)
<i>Patient characteristics</i>			
Male (%)	63.2	63.6	64.5
Mean age (years)	36.1	34.4	32.7
HIV positive (%)	26.8	26.8	23.3
<i>Disease characteristics</i>			
Extrapulmonary TB (%)	14.7	11.8	11.3
Bacteriologically confirmed (%)	–	75.3	70.4
Retreatment (%)	7.0	8.7	9.3
Drug resistant (%)	1.1	0.0	0.1
<i>Outcomes</i>			
Cured (%)	21.2	45.3	34.1
Treatment completed (%)	26.5	44.2	45.1
Died (%)	4.1	5.1	5.5
Treatment failed (%)	0.3	0.8	1.1
Lost to follow-up (%)	3.0	4.6	3.9

2 Data Quality Assurance: Outcome Discrepancies

2.1 Crosstab of TIBU vs. Paper Registry Outcomes

Table 2 shows the full cross-tabulation of treatment outcomes recorded in TIBU against outcomes recorded in the paper registry for all matched patients ($N = 3,370$). Cells are shaded to indicate the type of discrepancy:  Type I error (α): TIBU records success but paper records failure (false negative);  Type II error (β): TIBU records failure but paper records success (false positive);  incomplete data: no paper record present. Bold cells indicate the seven most common TIBU→paper correction pairs.

Table 2: Cross-tabulation of treatment outcomes in TIBU vs. paper registry. Bold = top 7 most common correction pairs.

Outcome in TIBU	Treatment Outcome in Paper Registry						Total
	Blank	C	D	F	LTFU	TC	
Blank	80 (2.4%)	22 (0.7%)	2 (0.1%)	2 (0.1%)	2 (0.1%)	12 (0.4%)	120
C	45 (1.3%)	1,032 (30.6%)	1 (0.0%)	3 (0.1%)	7 (0.2%)	60 (1.8%)	1,148
D	4 (0.1%)	3 (0.1%)	170 (5.0%)	0	1 (0.0%)	8 (0.2%)	186
F	0	1 (0.0%)	1 (0.0%)	28 (0.8%)	5 (0.1%)	2 (0.1%)	37
LTFU	3 (0.1%)	2 (0.1%)	1 (0.0%)	0	121 (3.6%)	6 (0.2%)	133
MT4	2 (0.1%)	3 (0.1%)	0	4 (0.1%)	1 (0.0%)	1 (0.0%)	12
NC	10 (0.3%)	34 (1.0%)	6 (0.2%)	0	4 (0.1%)	32 (0.9%)	87
TC	34 (1.0%)	81 (2.4%)	8 (0.2%)	0	13 (0.4%)	1,384 (41.1%)	1,520
TO	4 (0.1%)	43 (1.3%)	5 (0.1%)	0	13 (0.4%)	62 (1.8%)	127
Total	182	1,221	194	37	167	1,567	3,370

Color coding (bold = top 7 most common correction pairs):

 Type I error (α): TIBU success $\rightarrow$ paper failure (false negative): 0.9% of records

 Type II error (β): TIBU failure $\rightarrow$ paper success (false positive): 3.6% of records

 False missing (paper record absent): 5.4% of records

2.2 Error Rates by Patient Characteristics and Province

Table 3 reports false-negative, false-positive, and incomplete data rates broken down by patient characteristics (sex, age group, HIV status, disease characteristics) and province.

Table 3: DQA error rates by patient characteristics and province.

	N	False negative (%)	False positive (%)	Incomplete data (%)
<i>Sex</i>				
Male	1992	1.0	3.5	2.9
Female	1119	1.1	1.6	3.4
<i>Age group</i>				
<15	438	0.2	1.4	1.4
15–34	1418	1.3	2.9	3.0
35–54	929	0.8	3.0	3.7
55+	315	1.9	2.5	3.2
<i>HIV status</i>				
Positive	720	1.0	3.2	4.0
Negative	2330	1.0	2.7	2.8
<i>Disease characteristics</i>				
Extrapulmonary	351	0.6	2.0	3.7
Pulmonary	2754	1.1	2.9	2.9
Bacteriologically confirmed	2177	1.2	2.8	3.4
Not bacteriologically confirmed	934	0.6	2.8	2.2
Retreatment	290	1.7	1.7	3.1
New case	2755	0.9	2.9	2.4
<i>Province</i>				
Central	19	0.0	21.1	5.3
Coast	745	0.9	1.9	4.3
Eastern North	2	0.0	0.0	0.0
Eastern South	287	1.4	7.7	8.0
Nairobi	378	1.6	2.1	0.5
North Eastern	748	1.2	0.7	0.3
Nyanza North	394	0.5	2.3	1.0
Nyanza South	43	2.3	0.0	0.0
Rift Valley North	44	0.0	0.0	6.8
Rift Valley South	419	0.5	6.0	6.7
Western	32	3.1	3.1	3.1

2.3 Correction Lag

Table 4 summarises the number of days between the outcome date recorded in TIBU and the outcome date recorded in the paper registry, stratified by error type. A negative lag indicates the paper record was entered before TIBU.

Table 4: Days between TIBU and paper registry outcome dates, by error type.

Error type	N	Mean (days)	Median (days)	Q1 (days)	Q3 (days)
False negative	24	-65	-90	-140	0
False positive	19	52	43	2	130
False missing	96	No paper record — lag undefined			

Note: lag = paper outcome date – TIBU outcome date.

Negative lag indicates paper was recorded before TIBU.

2.4 Error Rates Over Time

Figure 1 plots DQA error rates across the study period.

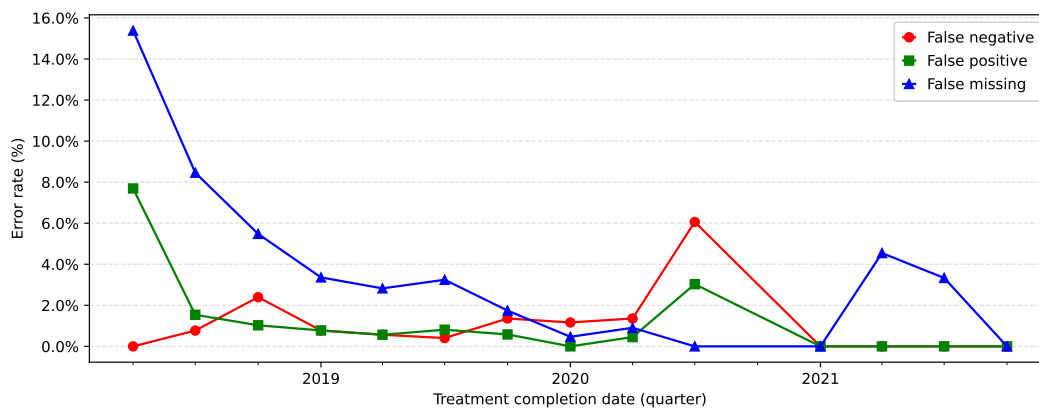


Figure 1: DQA error rates over time.